COURSE OUTLINE



DATA9102

Predictive Analytics Essentials

Fall 2020 - Current

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DATA9102 Predictive Analytics Essentials

COURSE DESCRIPTION

This course is an introductory course to predictive analytics including data interpretation, statistical testing, machine learning basics, data mining standards, and CRISP–DM. The focus of the course is on various types of statistical analysis to effectively identify patterns and capture insight from data and develop business solutions.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	24
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Describe statistics and how it is applied in the real world.	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
2	Calculate center and range of data	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
3	Describe distribution and sampling	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
4	Explain hypothesis testing and regression	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
5	Describe machine learning models	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)
6	Explain data mining: CRISP–DM and data visualizations for business	Thinking Skills (2), Digital Literacy (5) Numeracy and Financial (3) Communication (1)

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Introduction Module 1 – Statistic in Real World and Center and Range of the Data • What is statistics in the real world • Understand mean, median and mode • Understand standard deviation, range and outlier
Week 2	Module 1 – Statistic in Real World and Center and Range of the Data • What is statistics in the real world • Understand mean, median and mode • Understand standard deviation, range and outlier • Quiz 1
Week 3	Module 2 – Data Distribution and Sampling • What is normal distribution • Familiarize with probability and random variable • Understanding Sample size, central limit theorem and standard errors • Understanding confidence intervals
Week 4	Module 2 – Data Distribution and Sampling • What is normal distribution • Familiarize with probability and random variable • Understanding Sample size, central limit theorem and standard errors • Understanding confidence intervals
Week 5	Module 3 – Hypothesis Testing • Establishing a hypothesis test • Interpreting the significance of testing results • What is type I and type II error • Quiz 2
Week 6	Module 4 – Regression • What is regression • How to determine the fitting line • What is correlation coefficient • What coefficient of determination • Evaluating regression results
Week 7	Module 4 – Regression • What is regression • How to determine the fitting line • What is correlation coefficient • What coefficient of determination • Evaluating regression results
Week 8	Module 5 – Introduction to Machine Learning • What is supervised learning and applications • What is unsupervised learning and applications • What is a neural network models • Quiz 3
Week 9	Module 6 – Data Mining – CRISP DM • Understanding CRISP–DM for data mining • Introduction to Databases and their functionality • Learn how to desing queries and set up aggregation.
Week 10	Module 6 – Data Mining – CRISP DM • Understanding CRISP–DM for data mining • Introduction to Databases and their functionality • Learn how to design queries and set up aggregation
Week 11	Module 6 – Data Mining: CRISP–DM • Understanding CRISP–DM for data mining • Introduction to Databases and their functionality • Learn how to design queries and set up aggregation • Quiz 4
Week 12	Module 7 – Business Visualization Fundamentals • Understand charts and graphs • Understand how to read business charts • Determine what charts or graphs are best for different situations.
Week 13	Module 7 – Business Visualization Fundamentals • Understand charts and graphs • Understand how to read business charts • Determine what charts or graphs are best different situations. • Quiz 5
Week 14	Study week
Week 15	Study week & Final Exam

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2	Quiz 1 – Statistic in Real World and Center and Range of the Data	10%
3, 4	Quiz 2 – Data Distribution and Sampling and Hypothesis Testing	10%
4, 5	Quiz 3 – Regression and Introduction to Machine Learning	10%
6	Quiz 4 – Data Mining: CRISP-DM	10%
6	Quiz 5 – Business Visualization Fundamentals	10%
1, 2, 3, 4, 5, 6	Final Exam – Comprehensive Modules 1 to 7	50%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	Description	Value in GPA
P	Pass	N/A
F	Fail	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.